World Robot Olympiad 2025

SINGAPORE

FUTURE ENGINEERS

AI-STEAM SELF DRIVING CAR

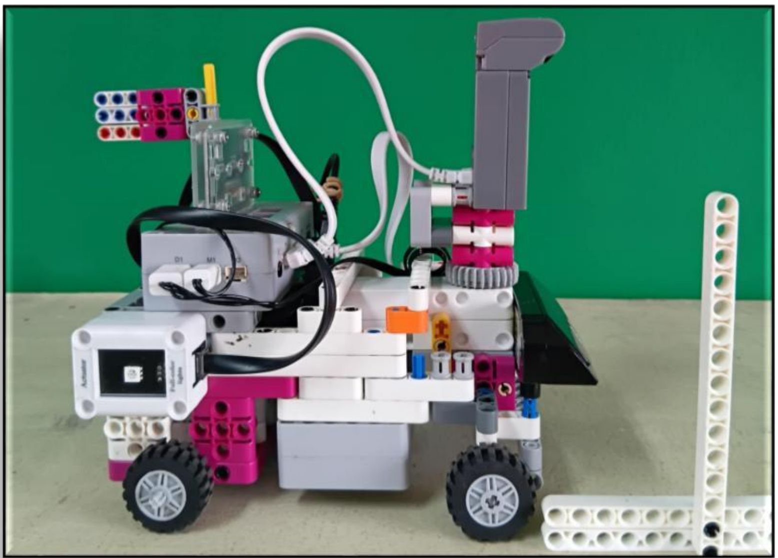
Team Name: SJWIS Stallions F.E. Team
Category: Future Engineers
Age Group: High School
Coach: Mr. Jimmy A. Melendres
Project Title: ATA 4.0 – Autonomous Self-Driving AI Robot
Platform: LEGO Technic + Nashenbot AI-STEAM
GitHub Repository: SJWISFE-Auto-car GitHub
Country: Philippines
Date: November 26-28, 2025

Introduction

The **ATA 4.0** (Autonomous Technic AI) is the fourth-generation self-driving robot developed by **Team Stallions** for the **Future Engineers | AI-STEAM Self-Driving Car Challenge**. Built on a LEGO Technic base and powered by **Nashenbot's AI-STEAM platform**, this project embodies a fusion of mechanical design, artificial intelligence, and sensor-driven automation.

Our primary objective was to design a compact, efficient, and intelligent autonomous vehicle capable of navigating complex obstacle environments and performing advanced decision-making using dual-camera vision. The robot integrates **two AI-enabled cameras**, a sound module, and a light-emitting module to enhance debugging and control. The core strategy focused on real-time color detection, adaptive pathfinding, dynamic camera angling, and wall detection logic.

ATA 4.0's design and programming process involved multiple engineering iterations, modular coding phases, and logic-based performance modules. This documentation outlines the robot's technical components, mobility systems, sensing and obstacle management strategies, as well as performance results and engineering innovations. Through this project, Team Stallions aimed to push the limits of vision-based robotics using creative AI integration and structured engineering methodology.

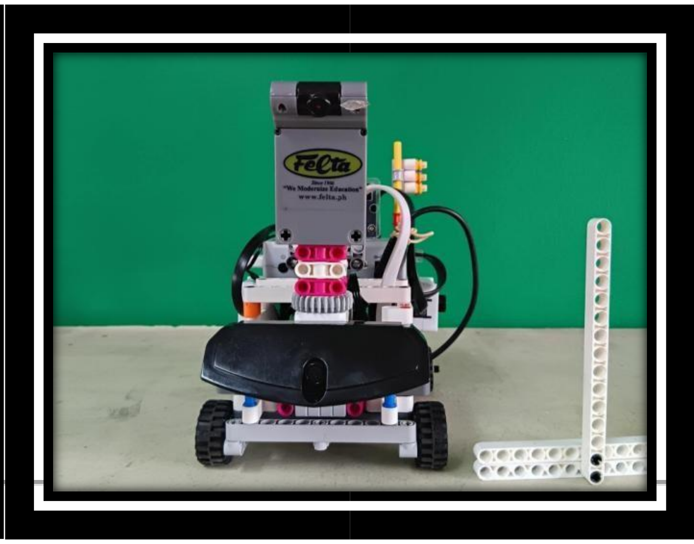


Mobility Management

Our robot, ATA 4.0, is designed with compact mobility in mind to efficiently navigate obstacle-heavy environments.

- **Drive System:** Dual-motor base using LEGO Technic components for precise left-right differential control.
- **Steering System:** Controlled via encoder-based servo motor adjusting the top camera for direction sensing.
- **Chassis Structure:** Lightweight, tightly packed frame using small wheels to enhance turning radius and reduce surface friction.
- **Camera Steering:** Servo-driven top camera at a 30–45° tilt for dynamic view angle, aiding in path correction and wall detection.

Image: Robot Side View and Top Vie w



Power and Sense Management –

- **Controller:** Nashenbot AI mainboard, supports modular sensors and camera input.
- **Cameras:**
  - **Top Camera**: For wall detection, line counting (blue/orange), and internal time process readings.
  - **Front Camera**: For color block detection (Red/Green), and following algorithm.
- **Sensors/Modules:**
  - **Light Module** and **Sound Module** for debug signaling.
- **Thresholds Calibration:** Before every match, values for blue, red, green, orange, and black are manually adjusted using `DEBUGMODE55` for real-time threshold calibration.

Obstacle Management –

- **Wall Detection:**
  - The **Top Camera** is used as a simulated distance sensor based on visual limitation.
  - When a wall is detected in view, robot counters with opposite turn direction to avoid collision.
- **Obstacle Challenge Strategy:**
  - Follows Red and Green blocks.
  - Red = Turn Right; Green = Turn Left, using the front camera.
  - Executes logic-based pathing using timers, camera angle control, and switching between turning modes (Turning vs Timed Mode).

Team Photo

GitHub Utilization

⚭ **GitHub Repo:** SJWISFE-Auto-car

Repository Contents:

- `/Schemes/` Folder: Includes camera configurations, calibration values, and setup schemes
  Source Files:
  - Setup Code (Variable Initialization, Thresholds, Servo Settings)
  - Phase 1 – Direction Initialization, Camera Angling
  - Phase 2 – Logic for Color Detection, Counting, Turning
- **SensorChange.txt:** Describes how the AI camera provides input for time processes and real-time decision making
- **Debug Mode Files:** Used for threshold visualization without external display

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Engineering Factor –

Key Innovations:

- **Modular Camera Logic:** Dual-camera system with distinct roles (Top for logic, Front for interaction).
- **Servo-Angle Encoding via `ReferenceValueEDTA`:** A workaround to control angle for motors without built-in angle values.
- **Time Process Variables:** Software-based timers to replace AIcode's wait function, allowing asynchronous event triggering.

- **Multi-phase Modular Programming:**
    - *Phase 1:* Direction, Camera Initialization
    - *Phase 2:* Movement, Color Logic, Timer Management, Display Feedback

Engineering Highlights:

- Custom logic for **Camera Angle State** and **InOutMode** for open/obstacle challenges
- Adaptive turning using two logical modes:
    - *Wall Detected → Counter-turn*
    - *No Wall → Turning Mode until detected again*
- Count Verification through *Pixel Color + Timer Countdown + Audio Feedback*

Conclusion

ATA 4.0 represents the culmination of iterative mechanical design, modular AI coding, and hardware-level debugging integration. Our focus on creating a **robust, adaptable, and vision-driven robot** enabled us to meet the challenges of both open and obstacle courses through teamwork, experimentation, and precise control.
